Q1.

The diagram shows part of the Periodic Table.

period																	
1							H										He
	I	II											III	IV	V	VI	VII	0
2	Li	**Be**											B	C	N	O	**F**	Ne
3	Na	**Mg**											Al	Si	P	S	**Cl**	Ar
4	K	**Ca**	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	**Br**	Kr
5	Rb	**Sr**	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	**I**	Xe

(a) Calcium burns brightly in oxygen, forming calcium oxide (CaO).
Calcium oxide reacts with water, forming a compound with the formula $Ca(OH)_2$.

(i) Give the name of the compound with the formula $Ca(OH)_2$.

……

1 mark

(ii) The compound, $Ca(OH)_2$, is slightly soluble in water.

Would you expect this solution to be **acidic**, **alkaline** or **neutral**?

……

1 mark

(b) The table below gives information about four compounds.
The molecules of each compound contain an atom of hydrogen and an atom of an element from group VII of the Periodic Table. The amount of energy needed to pull the two atoms apart is called the bond strength.

compound		bond strength in Kj/mol	action of heat on the compound
name	**formula**		
hydrogen fluride	HF	570	stable
hydrogen chloride	HCl	432	fairly stable
hydrogen bromide	HBr	366	some bromine formed
hydrogen iodide	HI	298	

Use the information in the table to answer the following questions.

(i) Suggest why hydrogen iodide may not exist at room temperature.

……

……

1 mark

(ii) Describe how the bond strength of these compounds varies in group VII.

……

……

1 mark

(iii) Which compound in the table requires the highest temperature to make it decompose?

……

1 mark

Maximum 5 marks

Q1.

(a) (i) calcium hydroxide

1

(ii) alkaline

accept 'strongly alkaline'

1

(b) (i) any **one** from

- because bond strength is less than that of HBr
  or less than 366 kJ/mol
  accept 'the bond is not strong enough'
- the bond strength is not enough to keep the atoms together
  accept 'the thermal energy is greater than the bond strength'
- bond strength decreases as you go down the group
  or as the period increases
- the compounds become increasingly unstable as you go down the group
  accept 'it is unstable'
  accept 'iodine is formed'

1

(ii) any **one** from

- it decreases from HF to HI **or** down the group
- it increases from HI to HF **or** up the group
  accept 'as the period goes up the strength goes down'

1

(iii) hydrogen fluoride **or** HF

1

[5]